

SORTING

Q1)

Question:

DS	Session	Sorting	Question Information	Level 1	Challenge 11
Problem Description: Tina owns a match making company, which even to her surprise is an extreme hit. She says that her success rate cannot be matched (Yes, letterplay!) in the entire match-making industry. She follows an extremely simple algorithm to determine if two people are matches for each other. Her algorithm is not at all complex, and makes no sense - not even to her. But she uses it anyway. Let's say that on a given day she decides to select n people - that is, n boys and n girls. She gets the list of n boys and n girls in a random order initially. Then, she arranges the list of girls in ascending order on the basis of their height and boys in descending order of their heights. A girl A_i can be matched to a boy on the same index only, that is, B_i and no one else. Likewise, a girl standing on A_k can be only matched to a boy on the same index B_k and no one else. Now to determine if the pair would make an ideal pair, she checks if the modulo of their heights is 0, i.e., $A_i \% B_i == 0$ or $B_i \% A_i == 0$. Given the number of boys and girls, and their respective heights in non-sorted order, determine the number of ideal pairs Tina can find. Constraints: $1 \leq \text{Test Cases} \leq 10^2$ $1 \leq N \leq 10^4$ $1 \leq A_i, B_i \leq 10^5$ Input Format: The first line contains number of test cases. Then, the next line contains an integer, n, saying the number of boys and girls. The next line contains the height of girls, followed by the height of boys. Output Format: Print the number of ideal pairs in a separate lines					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int t, n, i, j, temp, count;
5
6      printf("Input:\n");
7      scanf("%d", &t);
8
9      while (t--) {
10         scanf("%d", &n);
11
12         int g[10000], b[10000];
13
14         for (i = 0; i < n; i++)
15             scanf("%d", &g[i]);
16
17         for (i = 0; i < n; i++)
18             scanf("%d", &b[i]);
19
20         for (i = 0; i < n - 1; i++)
21             for (j = i + 1; j < n; j++)
22                 if (g[i] > g[j]) {
23                     temp = g[i];
24                     g[i] = g[j];
25                     g[j] = temp;
26                 }
27
28         for (i = 0; i < n - 1; i++)
29             for (j = i + 1; j < n; j++)
30                 if (b[i] < b[j]) {
31                     temp = b[i];
32                     b[i] = b[j];
33                     b[j] = temp;
34                 }
35
36         count = 0;
37
38         for (i = 0; i < n; i++)
39             if (g[i] % b[i] == 0 || b[i] % g[i] == 0)
40                 count++;
41
42         printf("%d\n", count);
43     }
44
45     return 0;
46 }
```

OUTPUT:

TEST CASE 1:

```
Input:
2
4
1 6 9 12
4 12 3 9
2
4
2 2 2 2
2 2 2 2
4
```

TEST CASE 2:

```
Input:
2
4
3 6 9 12
4 12 16 9
3
4
1 1 1 1
2 2 2 2
4
```